STUDY GUIDE: DESCRIPTION OF MOTION

Kinematics of a Material Point

Technical Context: Mechanics

Mechanics is the branch of physics studying the motion of objects.

- **Kinematics:** Describing motion (position, velocity, acceleration) without forces.
- **Dynamics:** Studying the forces that cause changes in motion.

I. Relativity of Motion

Motion is not absolute; it depends entirely on the chosen observer.

1. Frame of Reference

A complete Frame of Reference requires:

a) **Space Frame:** An origin O and a coordinate system (e.g., $(O, \vec{i}, \vec{j})$).

b) **Time Frame:** A clock with an origin of time ($t = 0$) and a unit (seconds).

2. Trajectory

The path followed by a moving object.

- **Rectilinear:** Straight line path.
- **Curvilinear:** Curved path (e.g., Circular or Parabolic).

II. Kinematics Variables

1. Position and Displacement

The position of a point M on a rectilinear axis is its abscissa $x = \overline{OM}$.

- **Displacement (Δx):** $x_{final} - x_{initial}$. This is a vector quantity (can be negative).
- **Distance (d):** The scalar length of the actual path ($d \geq 0$).

2. Velocity and Speed

- **Average Speed:** $v_{av} = \frac{d}{\Delta t}$
- **Instantaneous Velocity ($\vec{v}$):** The velocity at a specific time t, tangent to the trajectory in the direction of motion.

Standard Conversion

To move between units:

$$1 \text{ m/s} = 3.6 \text{ km/h} \quad \longleftrightarrow \quad 1 \text{ km/h} = \frac{1}{3.6} \text{ m/s}$$

III. Acceleration and Nature of Motion

Acceleration (a) measures the rate of change of velocity ($a = \frac{\Delta v}{\Delta t}$). Unit: m/s^2.

Condition	Kinematic Relationship	Nature of Motion
$v =$ constant	$a = 0$	Uniform Rectilinear (URM)
v and a same sign	$\vec{v} \cdot \vec{a} > 0$	Accelerated Rectilinear
v and a opposite signs	$\vec{v} \cdot \vec{a} < 0$	Decelerated Rectilinear

IV. Check for Understanding

Calculate the ***average acceleration*** *(a_{av}) of a vehicle that increases its speed from 10 m/s to 25 m/s over a duration of 5 seconds. State the nature of motion.*

Calculation Area